REVUB: User Manual (version 2.0.0)

Sebastian Sterl

Vrije Universiteit Brussel (VUB), Brussels, Belgium
Contact: sebastian.sterl@vub.be

Last updated: November 4, 2026



Contents

Note 1	Principles of REVUB	2
Note 2	Hydropower plant categorisation	2
Note 3	Simulations for “large” hydropower plants	4
Note 3.1	Conventional operation (for calibration)	4
Note 3.2	Balancing-oriented operation (dispatch)	6
Note 3.3	Reservoir shape	9
Note 3.4	Peaking suitability	10
Note 4	Simulations for “small” hydropower plants	10
Note 5	Optional pumped-storage assessment	12
Note 6	Modelling cascade plants	13
Note 7	Note on spinning reserve requirements	14
Note 8	Glossary	15
Note 9	Running the model: Input (A)	18
Note 10	Running the model: Core code (B)	20
Note 11	Running the model: Output graphics and tables (C)	20
Note 11.1	Results for individual power plants	21
Note 11.2	Results for multiple power plants	22
Note 12	Supplementary tips for model use	23

The REVUB model ("Renewable Electricity Variability, Upscaling and Balancing") was developed to model hydropower dispatch at high temporal detail, to investigate hydropower reservoir dynamics and drought resilience under different modes of operation, as well as to model hydro-backed integration of variable solar and wind power.

We first explain the general principles of the REVUB model (Note 1), based on which the different modules of the model are explained one by one: hydropower plant classification (Note 2); generation of calibration (guide) curves of reservoir dynamics (Note 3); and simulation and optimisation of hydropower dispatch with solar and wind power (Note 3 to Note 4). An optional, additional assessment of pumped-storage potential with REVUB is described in Note 5. The possibilities of modelling interactions between hydropower plants in cascades are described in Note 6. Some notes on spinning reserve requirements are summarised in Note 7. An overview table of terms and symbols used in the model description is given in Note 8.

The most recent version of the REVUB code was written for Python 3.11 and can be accessed via `https://github.com/VUB-HYDR/REVUB` and used under the MIT license. All equations given in this document are referenced in the code wherever relevant/used. The ways to prepare input, run the code, and interpret output are given in Note 9, Note 10 and Note 11. The manual ends with an overview of additional modelling tips in Note 12.

Note 1 Principles of REVUB

The basic idea of REVUB is to assess the potential of reservoir hydropower to assist in the regulation of river discharge, in the provision of flexibility to the power mix, and in supporting grid integration of variable renewable electricity (VRE) generation (solar PV and wind power). To do this, REVUB optimises reservoir operation from hourly to seasonal and multiannual time scales under different external circumstances.

REVUB calculates adapted rule curves required for hydropower dispatch in divergent scenarios—e.g. to ride through droughts, to meet a variable load, and/or to balance a given VRE portfolio[1–4]—and optimises the dispatch based on three criteria: (i) reservoir lake levels must align with "guide curves" of emptying-refilling cycles; (ii) the resulting hydropower operation, including co-optimised amounts of hydro-backed VRE, must be capable of consistently meeting a certain load at hourly, seasonal, and multiannual time scales; and (iii) downstream stable (e.g. environmental) flow constraints, as well as other technical limitations such as minimum turbine loads, must be respected.

An overview of the calculation steps performed by REVUB is given in Fig. S1. The steps outlined in this chart are described in more detail in the next sections, as indicated.

Note 2 Hydropower plant categorisation

In REVUB, hydropower plants are categorised into two groups, based on whether all natural inflow is allocated to regulation of outflow, or only a fraction of it, in other words: whether the reservoir capacity is deemed "large" or "small" compared to the inflow[i]. Generically, for plants in the "large" category, the idea would normally be to ensure outflow and power output are relatively consistent from month to month[5,6], enabling the plant to serve as baseload provider. Their rule curves often boil down to parameterisations of required outflow as a function of lake level[7]. For hydropower plants in the "small" category, rule curves should additionally take into account that certain fractions of the received water need to be directly released from the reservoir without being stored, for instance during the wet season(s), as reservoir capacity would be insufficient to store/regulate that component in a useful manner[8]. In practice, hydropower plants deemed "small" thus operate partly as run-of-river plants.

[i] A common benchmark for this would be whether the average natural inflow would take more or less than one year to fill the reservoir; but this is not necessarily applicable in every situation, as it depends on the seasonality of inflow too.

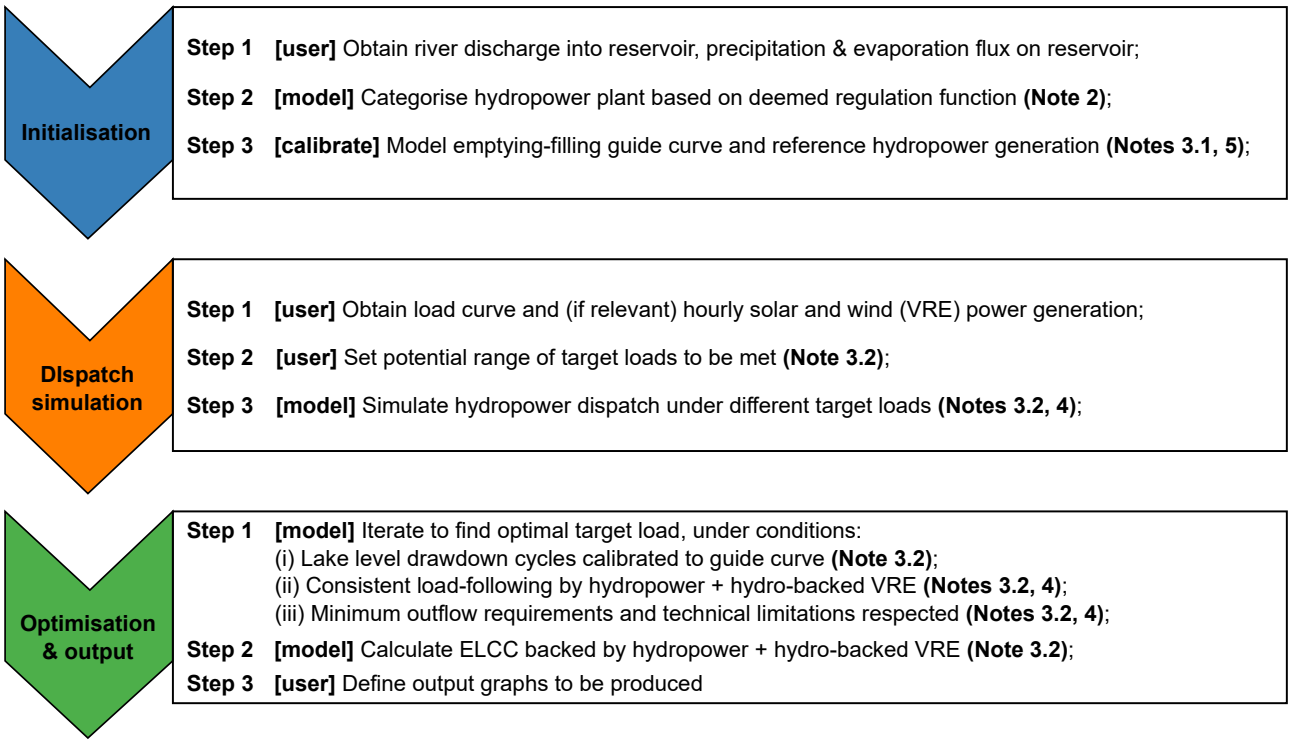


Figure S1: **Flowchart of the REVUB model approach**, explaining the steps taken in modelling hydro, solar and wind power and their integration. ELCC = Effective Load Carrying Capability. See also Note 9, Note 10 and Note 11 for the ways to prepare input, run the code, and interpret output.

In mathematical terms, knowing how much water a reservoir receives in comparison to its size can be done by calculating the dimensionless filling time τ_{fill}:

$$\tau_{fill} = \frac{V_{max}/T_{ref}}{\overline{Q_{in,nat}(t)}}, \tag{S1}$$

where V_{max} is the reservoir volume (in m^3), T_{ref} is the number of seconds in a reference period[ii], and $\overline{Q_{in,nat}(t)}$ denotes long-term average natural reservoir inflow (in m^3/s). Hydropower plants are classified by REVUB as large or small based on whether their τ_{fill} is larger or smaller than unity; the reference period can be changed by the user.

Regardless of whether a plant is classified as large or small—as long as it has regulating capacity, its operation could be adapted according to different system needs. For instance, the plant could be operated to provide baseload (using all regulation capacity to provide stable outflow), or to consistently ramp up and down to meet peak loads (using part of the regulation capacity to release more water during peak hours), or to provide balancing services to VRE (using part of the regulation capacity to release water during hours of low VRE availability). The latter two options require allocating a portion of the inflowing water for flexible use and releasing it to generate extra power whenever needed to meet a certain load.

In general, only part of the water intake can be allocated for flexible use, even for reservoirs with large storage capacities compared to the inflow. First, a minimum stable outflow usually has to be guaranteed for at least four reasons (two of them technical, and two environmental): (i) ensuring that turbines run in their high-efficiency range, which requires a certain minimum loading [*technical*]; (ii) safeguarding power system stability, since requirements for inertia impose having some synchronous capacity, such as from hydroturbine-driven generators, dispatched at all times[9] [*technical*]; (iii) safeguarding downstream ecological integrity through a minimum environmental outflow[10] [*environmental*]; and (iv)

[ii] For instance, the number of seconds in one year, in case we take a "one-year filling time" as reference to distinguish "large" and "small" plants.

supplying irrigation water demand downstream [*environmental*].
Thus, regulated reservoir outflow must have a stable component, and may additionally have a flexible component, which are denoted respectively $Q_{stable}(t)$ and $Q_{flexible}(t)$ (both in m^3/s). $Q_{stable}(t)$ can be further split up into a technical component $Q^{tech}_{stable}(t)$ and an environmental/irrigation component $Q^{env/irr}_{stable}(t)$. These components are not mutually exclusive, as one can contribute to the other; thus, in general, $Q_{stable}(t)$ is not the sum of both components, but simply the maximum value of both (the one that takes precedence over the other, thereby automatically fully covering the other).
Overall flexibility can be increased by reducing $Q_{stable}(t)$ in favour of $Q_{flexible}(t)$. Essentially, to allocate a certain amount of water for flexible use, the stable outflow will have to be reduced by a comparable amount such that the overall water budget remains unchanged. Since $Q^{env/irr}_{stable}(t)$ is usually prescribed externally, in practice, this means that allocating more water to flexibility implies reducing $Q^{tech}_{stable}(t)$, but that component also has lower limits determined largely by turbine characteristics.
Total reservoir outflow may have two other components: a seasonal ("run-of-river") component, denoted $Q_{RoR}(t)$, for hydropower plants deemed "small"; and an overflow-prevention component[7] $Q_{spill}(t)$, representing the release of water via spillways when filling levels approach critical values, e.g. during extremely wet periods. Thus, total reservoir outflow $Q_{out}(t)$ is, in general, given by

$$\begin{aligned} Q_{out}(t) &= Q_{stable}(t) + Q_{flexible}(t) + Q_{RoR}(t) + Q_{spill}(t) \\ &= \max\left[Q^{tech}_{stable}(t), Q^{env/irr}_{stable}(t)\right] + Q_{flexible}(t) + Q_{RoR}(t) + Q_{spill}(t). \end{aligned} \tag{S2}$$

for both "large" and "small" hydropower plants. Correspondingly, hydropower generation consists of components $P^{hydro}_{stable}(t)$, $P^{hydro}_{flexible}(t)$ and $P^{hydro}_{RoR}(t)$ (in MW) [iii].
In the next sections, the simulations in REVUB of flexibility provision by "large" and "small" hydropower plants are discussed in detail. Note 3 and Note 3.4 deal with "large" plants—which are the exception rather than the rule, but starting with these is informative as the mathematics are somewhat simpler; Note 4 generalises the principles to "small" plants.

Note 3 Simulations for "large" hydropower plants

The volume of water $V(t)$ contained in a large hydropower plant's reservoir is given by the recursive relation

$$V(t+\Delta t) = V(t) + \left[Q_{in,nat}(t) - Q_{stable}(t) - Q_{flexible}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)]\,\frac{A(t)}{\rho}\right]\Delta t, \tag{S3}$$

where $\pi(t)$ and $\lambda(t)$ are respectively precipitation and evaporation flux ($kg/m^2/s$); $A(t)$ is the surface area of the reservoir lake (m^2); ρ is the density of water (kg/m^3); and Δt is the simulation time step (one hour is the default in REVUB). Seepage losses, which are usually very small in reservoirs[8,11], are neglected in the REVUB code (but could be included e.g. via a correction factor to total inflow). A schematic diagram of the involved water fluxes is given in Fig. S2.
Note 3.1 explains how $Q_{stable}(t)$ and the corresponding power generation $P^{hydro}_{stable}(t)$ are modelled for baseload-oriented operation, in which the flexible components are zero. Simulating such operation is necessary to have a reference time series of long-term lake volume fluctuations—a guide curve—needed (i) to calibrate the model to observations (if available), and (ii) as optimisation criterion in later model runs on flexible hydropower dispatch. Subsequently, Note 3.2 explains how $Q_{stable}(t)$ and $Q_{flexible}(t)$, and the corresponding power generation components $P^{hydro}_{stable}(t)$ and $P^{hydro}_{flexible}(t)$, are modelled for balancing-oriented operation, i.e. dispatch runs.

Note 3.1 Conventional operation (for calibration)

We denote conventional, baseload-oriented operational rules with the abbreviation "CONV". By definition, $Q_{flexible,CONV}(t)$ and $P^{hydro}_{flexible,CONV}(t)$ are zero. The default way of modelling $Q_{stable,CONV}(t)$

[iii] It is assumed that overflow prevention releases are never used for power generation, to avoid damaging turbine equipment and because such high flows would normally exceed turbine capacity anyway.

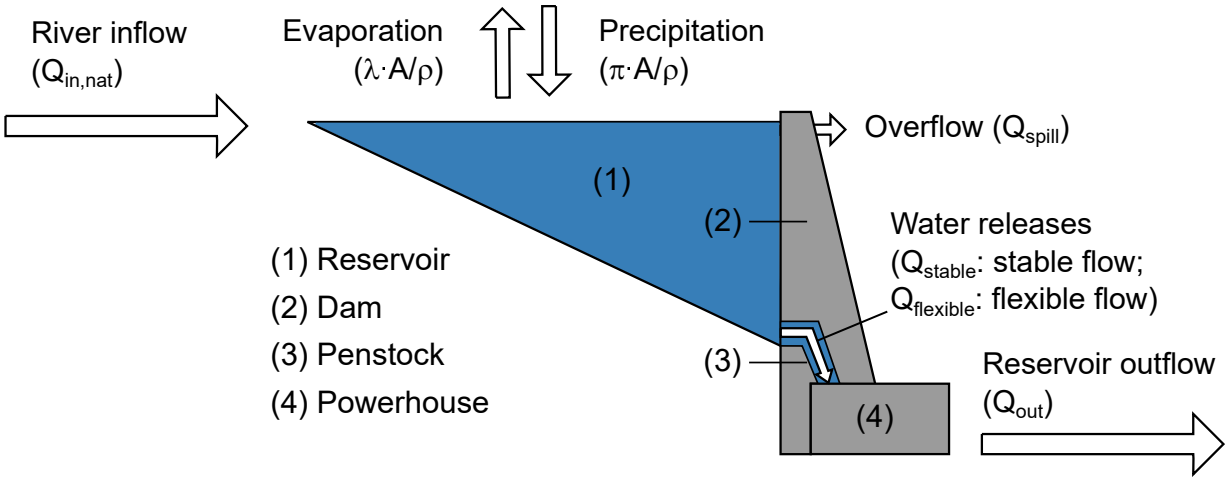


Figure S2: **Flowchart of the water gains and losses of a large hydropower plant as modelled in REVUB.** Gains consist of river discharge and lake surface precipitation; losses consist of reservoir outflows and lake surface evaporation.

in REVUB is based on logarithmic-exponential release rules[6,12], which works well in case conventional operational rules are unknown, and can be determined on the basis of long-term average inflow $\overline{Q_{in,nat}(t)}$ [iv]. Under the assumption that environmental/irrigation outflow $Q_{stable}^{env/irr}(t)$ is prescribed at all times, approximate rules used in REVUB are:

$$Q_{stable,CONV}^{tech}(t) = \left\{ \begin{array}{ll} \overline{Q_{in,nat}(t)} \left[d_{min} + \ln\left(\kappa \left[\frac{V_{CONV}(t)}{V_{max}} \right]^{\phi} + 1 \right) \right], & \text{for } V_{CONV}(t)/V_{max} < f_{opt} \\ \overline{Q_{in,nat}(t)} \left[\exp\left(\gamma \left[\frac{V_{CONV}(t)}{V_{max}} - f_{opt} \right]^{2} \right) \right], & \text{for } V_{CONV}(t)/V_{max} \geq f_{opt} \end{array} \right\}, \quad \text{(S4)}$$

where d_{min} is the fraction of yearly average inflow required as minimum stable outflow; V_{max} is the maximum reservoir storage capacity; f_{opt} is the optimal filling fraction; and κ, ϕ and γ are constants. κ and ϕ are given[6] by

$$\kappa = f_{opt}^{-\phi} \left[\exp\left(1 - d_{min}\right) - 1 \right], \quad \text{(S5)}$$

and

$$\phi = \alpha \tau_{fill}^{1/2}. \quad \text{(S6)}$$

Recommended default values are $f_{opt} = 80\%$, $\alpha = 2/3$, and $\gamma = 10$ based on generalised reservoir operation rules[6,12]; for d_{min}, values may be chosen based on requirements for minimum turbined flow to prevent turbines running at low efficiency, and/or generalised environmental flow rules[10] (the former will usually take precedence over the latter). It is possible (and arguably desirable when running scenarios for the first time) for the user to not specify a value for d_{min}, and instead provide a value of minimum required load f_{min} on one single turbine (expressed as fraction of maximum turbine throughflow), reflecting the lower bound of the turbine's operating range, based on which REVUB will calculate and work with a default value of d_{min} [v]. All the preceding parameters can be changed by the user in the process of calibration to historical data.

The overflow prevention component $Q_{spill,CONV}$ is modelled as

$$Q_{spill,CONV}(t) = \left\{ \begin{array}{ll} 0, & \text{for } V_{CONV}(t)/V_{max} < f_{spill} \\ \max\left[0, \left(Q_{in,nat}(t) + [\pi(t) - \lambda(t)] \frac{A_{CONV}(t)}{\rho} \right)(1+\mu) - Q_{stable,CONV}(t) \right], & \text{for } V_{CONV}(t)/V_{max} \geq f_{spill} \end{array} \right\}, \quad \text{(S7)}$$

[iv] By default, REVUB considers this average to be across the entire modelled period, but the user can alternatively select a specific part of the input period on which to base the average.

[v] The calculation is as follows: $d_{min}^{default} = (f_{min} Q_{turb}^{max} / N_{turbines}) / \overline{Q_{in,nat}(t)}$, where $N_{turbines}$ is the number of turbines (units) of the plant. The term $f_{min} Q_{turb}^{max} / N_{turbines}$ represents the minimum flow in m^3/s required to keep one turbine running within its operating range.

where f_{spill} is the filling fraction at which the overflow prevention starts[vi], and μ represents a small fraction such that lake levels are brought below critical levels $f_{spill}V_{max}$ as quickly as possible without releasing unnaturally high flows downstream; recommended values are $f_{spill} = 95\%$ and $\mu = 0.1$, but these can be changed by the user.

To simulate minimum drawdown levels when facing drought-like situations, one further rule is added to equation (S4): outflow, and with it hydropower production, is automatically curtailed [7,13] if the volume levels $V_{CONV}(t)$ dip below critical levels $f_{stop}V_{max}$, and only restarted once volumes have recovered to $f_{restart}V_{max}$. The values f_{stop} and $f_{restart}$ must be specified by the user, ideally based on reservoir bathymetry and knowledge of minimum operational levels (see Note 3.3). The two values can, of course, be set equal to one another.

The corresponding stable power output $P^{hydro}_{stable,CONV}(t)$ can then be calculated as

$$P^{hydro}_{stable,CONV}(t) = \min\left[Q_{stable,CONV}(t); Q^{max}_{turb}\right]\eta_{turb}\rho g h_{CONV}(t), \tag{S8}$$

where η_{turb} is the turbine efficiency (%)[vii], g the gravitational acceleration (9.81 m/s^2), and $h(t)$ the hydraulic head (m), i.e. the difference in water level between the headwater behind the dam and the tailwater at the turbines. The value of η_{turb} should be user-defined, ideally based on knowledge of the plant's turbines[viii]. The calculation of $h(t)$ is explained in Note 3.3. Q^{max}_{turb} is the maximum turbine throughput (at which the power generating capacity of the turbine is fully used). If this value cannot be found, it can be approximated with $Q^{max}_{turb} \approx P^{r}_{turb}/(\eta_{turb}\rho g h_{max})$, where P^{r}_{turb} is the rated power capacity of the hydropower plant and h_{max} is the maximum hydraulic head.

With these rules, results are mostly independent of arbitrary initial conditions (convergence to the same time series happens typically within two simulation months). REVUB uses the initial condition $V_{CONV}(0) = f_{init}V_{max}$ (and the corresponding lake area and water level; see Note 3.3) where the user can specify f_{init}.

We note that, in case the user wishes to include their own conventional reservoir rules instead of equation (S4), this is possible but would require the user adapting the REVUB code and changing the CONV simulation function.

Note 3.2 Balancing-oriented operation (dispatch)

Reservoir dispatch oriented towards meeting variable load and balancing VRE, is denoted with the abbreviation "BAL". For reservoirs of large hydropower plants, BAL operation is modelled as follows[ix]. First, the needs for sub-daily to seasonal dispatching patterns must be established. The load difference $P_d(t)$ between total inflexible power generation (stable hydropower, solar power, and wind power) and power demand is calculated as:

$$\begin{aligned} P_d(t) &= P^{hydro}_{stable,BAL}(t) + c_{solar} \cdot CF_{solar}(t) + c_{wind} \cdot CF_{wind}(t) - L(t) \\ &= P_{inflexible}(t) - L(t). \end{aligned} \tag{S9}$$

Here, c_{solar} and c_{wind} represent a certain amount of solar PV and wind power capacity, respectively (in MW), whose value is to be co-optimised by REVUB alongside the dispatch optimisation; $CF_{solar}(t)$

[vi] Note that every hydropower plant has a certain safety level, i.e. the dam is always somewhat higher than the maximum water level [7], to mitigate flood risk. Because of this safety level, should the overflow procedure fail for any reason, the dam would not immediately overflow. However, flooding would occur along the shores of the lake, which is why it is always prudent to prevent straining the safety level.

[vii] Hydropower plants usually produce reactive power next to active power (i.e. the power factor is less than unity). For purposes of REVUB, representing the power factor can be done through this same efficiency factor. For instance, to represent a conversion efficiency of 95% (a turbine quantity) and a power factor of 80% (an electric circuit quantity), one could set $\eta = 95\% \times 80\% = 76\%$.

[viii] The efficiency of hydroturbines depends on the effective turbined flow at each moment, but is typically above 80% in realistic ranges of the latter [14,15]. As explained above, the value d_{min} should at least reflect the fact that at any given moment, at least one of the $N_{turbines}$ turbines of a given plant must be active and running at a certain minimum partial load to allow consistent operation at high efficiency.

[ix] We provide here the generic approach for flexible operation in the presence of solar and wind power. For dispatch runs without VRE, i.e. aimed at meeting a variable load with hydropower only, the mathematics are largely identical; see Note 12.

and $CF_{wind}(t)$ represent the solar and wind capacity factor for each time step, respectively; and $L(t)$ represents a certain target load profile that is to be followed[x].

The time series $L(t)$, by default, is set as follows:

$$L(t) = P_{load} L_{norm}(t), \tag{S10}$$

with P_{load} a constant determining the average load to be followed (and reflecting the constraints on overproduction, see below), and $L_{norm}(t)$ the normalised ($\overline{L_{norm}(t)} = 1$) hour-to-hour load profile. P_{load} is calculated as

$$P_{load} = p_i(P_{inflexible}(t)), \tag{S11}$$

where $p_i(x)$ denotes the ith percentile of a variable x. Here, i thus represents the percentile of $P_{inflexible}$ not exceeding P_{load}; in other words, the percentage of time during which RE generation should not exceed the average load to be followed. Generally, the lower i, the higher the amount of allowed excess production (overproduction) as compared to $L(t)$.

In REVUB, the time series $CF_{solar}(t)$ and $CF_{wind}(t)$ must be provided as model input by the user. These could represent power generation from single locations, or weighted averages across the locations for on-grid solar or wind power selected for the region under scrutiny, to simulate the feed-in of solar and wind power from various locations into the same power grid. In case of the latter, we recommend assuming that the total capacity is distributed across locations within the region according to site-specific weight factors w^n_{solar} and w^n_{wind}. Then, $CF_{solar}(t)$ and $CF_{wind}(t)$ would be given by

$$CF_{solar}(t) = \frac{\sum_{n=1}^{N_{solar}} CF^n_{solar}(t) w^n_{solar}}{\sum_{n=1}^{N_{solar}} w^n_{solar}};\ CF_{wind}(t) = \frac{\sum_{n=1}^{N_{wind}} CF^n_{wind}(t) w^n_{wind}}{\sum_{n=1}^{N_{wind}} w^n_{wind}}, \tag{S12}$$

with $CF^n_{solar}(t)$ and $CF^n_{wind}(t)$ the capacity factors of solar PV or wind power for each time step, respectively, at each individual site with index n; and N_{solar} and N_{wind} the number of sites for solar PV or wind power generation, respectively. The weight factors w^n_{solar} and w^n_{wind} represent preferences for certain sites over others for the development of solar PV and wind parks. To determine the site-specific weight factors, one could take e.g. the available area for deployment in each different site and assume it scales linearly with potential capacity deployment, as per ref. [16]. Another approach is that of ref. [17], which took a site's weight factor to be equal to the multi-year average capacity factor for solar PV or wind power at that site, reflecting a preference for sites with higher yield:

$$w^n_{solar} = \overline{CF^n_{solar}(t)};\ w^n_{wind} = \overline{CF^n_{wind}(t)}. \tag{S13}$$

The capacity ratio $c_{solar} : c_{wind}$ should also be user-defined; it represents the relative share of solar and wind capacity to be deployed by the model. Sensitivity tests to find an optimal ratio can then be performed by running a range of scenario variations, using different values of the capacity ratio.

Second, in BAL operation, $Q_{stable}(t)$ is reduced in favour of $Q_{flexible}(t)$ such that $L(t)$ can be met by the combination of stable hydropower, flexible hydropower, solar power and wind power. In the REVUB code, the default approach (which could be adapted by the user in the code) is to take $Q^{tech}_{stable,BAL}(t)$ as a fraction of the average inflow[xi]:

$$Q^{tech}_{stable,BAL}(t) = (1 - C_{OR})\overline{Q_{in,nat}(t)}. \tag{S14}$$

Here, the fraction C_{OR} denotes the "operating reserve coefficient" [18], and determines the amount of water available for flexible use. Theoretically, as $C_{OR} \to 1$, the hydropower plant in question would operate near-completely flexibly and provide almost no minimum stable load, beyond the "coincidental" one from the environmental and irrigation outflow. This would also imply that the outflow of the dam could become extremely low for prolonged periods; for instance, when solar/wind power

[x] Note that for the terms in equation (S9), MW and MWh are interchangeable units, due to the hourly time step employed in REVUB. Similarly, CF could be interpreted both as actual power generation divided by maximum potential generation in each hour (dimensionless fraction), or as power generation per unit of capacity (in hours).

[xi] As for the CONV rules, by default this average is taken across the entire modelling period, but the user can alternatively select a part of that period as basis for the average.

generation is high and/or demand is low. Such operation would likely violate inertia requirements and minimum load constraints of the plant. As default, REVUB therefore uses an upper bound of $C_{OR}^{max} = 1 - d_{min}$, such that the minimum outflow under BAL is always consistent with that under CONV. The selection of the optimal C_{OR} for each hydropower plant is explained in Note 3.4.

$P_{stable,BAL}^{hydro}(t)$ is calculated analogously to $P_{stable,CONV}^{hydro}(t)$ in equation (S8):

$$P_{stable,BAL}^{hydro}(t) = \min\left[Q_{stable,BAL}(t); Q_{turb}^{max}\right]\eta_{turb}\rho g h_{BAL}(t). \tag{S15}$$

Third, the necessary amount of flexible outflow $Q_{flexible,BAL}(t)$ and corresponding flexibly produced power $P_{flexible,BAL}^{hydro}(t)$ are determined for the instances when $P_{inflexible}(t)$ cannot meet $L(t)$, i.e. when $P_d(t) < 0$. The following rule then applies for $P_{flexible,BAL}^{hydro}(t)$ to maximise the followed load under peak capacity constraints and ramping constraints[19]:

$$P_{flexible,BAL}^{hydro}(t) = \left\{ \begin{array}{ll} 0, & \text{for } P_d(t) \geq 0 \\ \min\left[Q_{turb,flexible}^{pot}(t)\eta_{turb}\rho g h_{BAL}(t); \min\left[|P_d(t)|; P_{flexible,BAL}^{hydro}(t-1) + \Delta P_{turb}^{ramp}\right]\right], & \text{for } P_d(t) < 0 \;\&\; \Delta P_d(t) < 0 \\ \min\left[Q_{turb,flexible}^{pot}(t)\eta_{turb}\rho g h_{BAL}(t); \max\left[|P_d(t)|; P_{flexible,BAL}^{hydro}(t-1) - \Delta P_{turb}^{ramp}\right]\right], & \text{for } P_d(t) < 0 \;\&\; \Delta P_d(t) \geq 0 \end{array} \right\}, \tag{S16}$$

where $\Delta P_d(t) = P_d(t) - P_d(t-1)$ determines whether the hydropower plant should ramp up ($\Delta P_d(t) < 0$) or down ($\Delta P_d(t) \geq 0$); ΔP_{turb}^{ramp} is the maximum ramp rate of the plant (in MW per time step); and

$$Q_{turb,flexible}^{pot}(t) = \max\left[0; Q_{turb}^{max} - Q_{stable,BAL}(t); \frac{P_{turb}^{r} - P_{stable,BAL}^{hydro}}{\eta_{turb}\rho g h_{BAL}(t)}\right]. \tag{S17}$$

This gives a flexible outflow

$$Q_{flexible,BAL}(t) = \frac{P_{flexible,BAL}^{hydro}(t)}{\eta_{turb}\rho g h_{BAL}(t)}. \tag{S18}$$

Fourth, the overflow prevention component $Q_{spill,BAL}(t)$ is calculated analogously to equation (S7):

$$Q_{spill,BAL}(t) = \left\{ \begin{array}{ll} 0, & \text{for } V_{BAL}(t)/V_{max} < f_{spill} \\ \max\left[0, \left(Q_{in,nat}(t) + [\pi(t) - \lambda(t)]\frac{A_{BAL}(t)}{\rho}\right)(1+\mu) - Q_{stable,BAL}(t) - Q_{flexible,BAL}(t)\right], & \text{for } V_{BAL}(t)/V_{max} \geq f_{spill} \end{array} \right\}. \tag{S19}$$

As in CONV, to ensure that lake levels do not dip precariously low, $Q_{stable,BAL}$ and $Q_{flexible,BAL}$ are reduced to zero when $V_{BAL}(t) < f_{stop}V_{max}$ and restarted once $V_{BAL}(t) \geq f_{restart}V_{max}$ [xii].

Since one needs the hydraulic head $h_{BAL}(t)$ to calculate $P_{stable,BAL}^{hydro}(t)$ from equation (S15), but $h_{BAL}(t)$ follows only from the overall operation, which requires knowledge of $P_{stable,BAL}^{hydro}(t)$ to calculate $P_d(t)$ according to equation (S9), this problem is solved iteratively by REVUB with an initial guess of $P_{stable,BAL}^{hydro}(t) = (1 - C_{OR})P_{stable,CONV}^{hydro}(t)$, repeated until convergence (typically, 2-3 iterations suffice). The calculation of $h(t)$ is the same as for CONV and is explained in Note 3.3.

At given C_{OR} and given ratio $c_{solar} : c_{wind}$, REVUB optimises the above operation by identifying the time series $L(t) = L_{opt}(t)$, and the corresponding optimal $c_{solar} = c_{solar}^{opt}$ and $c_{wind} = c_{wind}^{opt}$, for which the resulting lake level time series is most comparable to what it would be under CONV operation. This optimisation is performed by finding the minimum of the relative deviation Ψ between the CONV and BAL lake levels across all time steps of an entire simulation period[xiii], as a function of c_{solar} and c_{wind}:

$$\Psi(c_{solar}, c_{wind}) = \frac{\overline{|(V_{BAL}(t) - V_{CONV}(t)|}}{\overline{V_{CONV}(t)}}; \tag{S20}$$

where

$$\min(\Psi) = \Psi(c_{solar}^{opt}, c_{wind}^{opt}). \tag{S21}$$

[xii] The length of periods when hydropower production is forced to stop, is important for due diligence planning and reflects in parameters such as the $P_{guaranteed}^{90}$ power (MW), i.e. the power that is guaranteed 90% of the time. Integration of VRE will generally improve such parameters, since VRE can continue producing even when hydropower is temporarily stopped and despite the fact that the flexibility from hydropower is lost in those periods. REVUB calculates $P_{guaranteed}^{xx}$ for a user-defined percentile xx for both CONV and BAL scenarios to showcase this.

[xiii] Or a user-selected part of that simulation period, in line with the CONV and BAL rules.

Note that the code could also be adapted to optimise lake levels compared to a certain mathematical rule curve, instead of to the outcome of "conventional" operational rules (Note 3.1). This would simply require exchanging $V_{CONV}(t)$ in equation (S20) by the corresponding rule curve of lake volume. Note furthermore that even when doing this, it is still useful to calculate the outcomes (hydropower generation and lake levels) resulting from conventional reservoir management, to verify (i) how well the rule curve can be followed *in general* under the given hydroclimate and occurrence of wet/dry years, and (ii) the extent to which the outcomes of flexible operation would differ from baseload-oriented operation.

REVUB allows the user the option to ensure that power droughts resulting from $V(t)$ dipping lower than $f_{stop}V_{max}$ can never be more severe under BAL operation than under CONV, by automatically discarding any solutions to equation (S21) where this would be the case.

Once the optimal solution is found, the Effective Load Carrying Capability (ELCC) of the hydro-solar-wind mix is calculated as follows. The maximum followable load $L_{followed}(t) = P_{followed}L_{norm}(t)$, which the hydro-solar-wind mix can meet without any load loss, is identified: this is the load at which the residual load ($L_{res}(t)$; the load minus the generation) has a maximum equal to a user-defined Loss of Energy Expectation (LOEE). The default is LOEE = 0, but this can be relaxed by the user. Thus, $P_{followed}$ is the value for which $\max[L_{res}(t)] = \text{LOEE}$, with $L_{res}(t)$ defined as:

$$L_{res}(t) = P_{followed}L_{norm}(t) - \left[P^{hydro}_{stable,BAL}(t) + P^{hydro}_{flexible,BAL}(t) + c^{opt}_{solar} \cdot CF_{solar}(t) + c^{opt}_{wind} \cdot CF_{wind}(t)\right]. \quad \text{(S22)}$$

The ELCC is then defined as the integral of $L_{followed}(t)$ over all time steps in a simulation year:

$$\text{ELCC} = \int_{year} L_{followed}(t)dt. \quad \text{(S23)}$$

The ELCC thus represents the total yearly load followed by each hydropower plant in combination with the solar and wind power whose variability it can compensate, under the optimal solution. The aggregate of ELCC across all hydropower plants represents the total followable load when all hydropower plants optimally contribute to compensating solar-wind variability; it is abbreviated ELCC_{tot} hereafter.

Ideally, $L_{followed}(t)$ should be equal to $L_{opt}(t)$ (and it usually is), but due to peaking constraints of hydropower plants (eq. (S17)), this is not guaranteed in every case. When $L_{followed}(t) < L_{opt}(t)$, the power plant's peaking capabilities are insufficient to meet all required peaks in $L_{opt}(t)$ together with solar and wind power. In such cases, the hydropower plant should run at somewhat lower flexibility, i.e. lower C_{OR}, at which a lower $L_{opt}(t)$ will apply that would put less strain on the plant's peaking capabilities, ensuring that $L_{followed}(t) \approx L_{opt}(t)$. This resimulation is done automatically by the REVUB code (Note 3.4).

The total contributions of hydro, solar and wind power to yearly electricity generation, denoted respectively $E^{hydro}_{reservoir}$, E_{solar} and E_{wind}, are obtained by integrating their respective power output across all time steps in a simulation year. First, $E^{hydro}_{reservoir}$ is equal to

$$E^{hydro}_{reservoir} = E^{hydro}_{stable} + E^{hydro}_{flexible} = \int_{year} \left(P^{hydro}_{stable}(t) + P^{hydro}_{flexible}(t)\right)\, dt. \quad \text{(S24)}$$

Second, E_{solar} and E_{wind} are obtained by multiplying c^{opt}_{solar} and c^{opt}_{wind} by $CF_{solar}(t)$ and $CF_{wind}(t)$, respectively, and integrating:

$$\begin{aligned} E_{solar} + E_{wind} &= \int_{year} \big(P_{solar}(t) + P_{wind}(t)\big)dt \\ &= \int_{year} \big(c^{opt}_{solar} \cdot CF_{solar}(t) + c^{opt}_{wind} \cdot CF_{wind}(t)\big)dt. \end{aligned} \quad \text{(S25)}$$

Note 3.3 Reservoir shape

Calculating the hydraulic head $h(t)$ for each time step requires knowing the bathymetry and volume-area-depth relationship of each reservoir. Typical curves approximating such relationships need to be calibrated on a reservoir-to-reservoir basis. The user should provide bathymetric curves as input to

each simulation. In case these are not available, we recommend using an archetypal reservoir shape function[20,21], modelling reservoirs' area-volume relationships with the equation

$$A(t) = A_{max}\left[\frac{V(t)}{V_{max}}\right]^{(1-C_{reservoir})}, \tag{S26}$$

where $C_{reservoir} = V_{max}/(A_{max}z_{max})$ is the "reservoir coefficient", and A_{max} and z_{max} represent the maximum reservoir lake area and depth, respectively.
For any $V(t+\Delta t)$ calculated using equation (S3), the corresponding $A(t+\Delta t)$ can then be calculated using equation (S26). To obtain the hydraulic head $h(t+\Delta t)$ at each time step, the incremental change $\Delta h(t)$ is added to the head $h(t)$ of the previous time step, using the first-order approximation

$$\Delta h(t) \approx \frac{V(t+1)-V(t)}{A(t)}. \tag{S27}$$

When using equation (S26), before any simulation according to CONV or BAL in REVUB, a dummy simulation spanning all possible values of V, and thus of A and h, should be run to obtain a calibration (h, V) curve. In CONV and BAL simulations, the head $h(t+\Delta t)$ can then always be obtained from this calibration curve after $V(t+\Delta t)$ has been determined from equation (S3).

Note 3.4 Peaking suitability

Since hydropower plants differ in terms of peaking capabilities, operational strategies must be selected with care. If hydropower plants are operated at a too high C_{OR}, the corresponding peaks in flexible water release may exceed the maximum turbine throughput on a structural basis (cf. equation (S17)), meaning that the plant will be structurally unable to meet peak demand as well as leading to loss of spinning reserves (see Note 7). Typically, this would first occur in the seasons with highest peak demand. Therefore, each hydropower plant's operation should happen at an optimised value $C_{OR} = C_{OR}^{opt}$ that ensures this is not the case, while maintaining adequate levels of flexibility. In REVUB, the default procedure is to define the turbine utilisation rate $k_{turb}(t)$ at each time step:

$$k_{turb}(t) = \frac{Q_{stable}(t) + Q_{flexible}(t)}{Q_{turb}^{max}}, \tag{S28}$$

and define hydropower plant operation (at given C_{OR}) as unsuitable for peaking purposes when $p_j(k_{turb})$, i.e. the jth percentile of k_{turb}, is unity; in other words, when turbine capacity is fully exhausted in at least $100-j$ out of every 100 hours (for instance, if $j = 99$, then the criterion $p_{99}(k_{turb}) = 1$ means that turbine capacity is fully exhausted in at least 1 out of 100 hours). The value of j can be changed by the user.
As default for each hydropower plant, REVUB uses $C_{OR}^{opt} = C_{OR}^{max}$, the maximum allowed operational flexibility. If operation is found to be unsuitable for peaking purposes at C_{OR}^{max}, the REVUB code automatically resimulates with incrementally reduced C_{OR}, until a value C_{OR}^{opt} is identified for which $p_{99}(k_{turb}) < 1$.

Note 4 Simulations for "small" hydropower plants

To assess the balancing potential of small hydropower plants (those with less than a year of storage), which are in reality much more ubiquitous than large hydropower plants, we define the alternative filling time $\tau_{fill,frac}$ corresponding to the amount of years it would take for a fixed fraction of the incoming flow, denoted $Q_{in,frac}(t)$, to fill the reservoir:

$$\tau_{fill,frac} = \frac{V_{max}/T_{ref}}{\overline{Q_{in,frac}(t)}}, \tag{S29}$$

where $Q_{in,frac}(t)$ is defined as

$$Q_{in,frac}(t) = f_{reg}Q_{in,nat}(t), \tag{S30}$$

with f_{reg} a suitable fraction, representing the fraction of incoming water available for storage and, therefore, flexible use. This fraction can be input in REVUB by the user. In case the user is unsure

of a pertinent value, it can be left empty and REVUB defaults to a standard value determined by solving $\tau_{fill,frac} = 1$ with a reference period of one year; that is, f_{reg} then represents the fraction of the incoming water that would take one year to fill the reservoir on average. Note that this default is very realistic for hydropower plants on rivers with extremely seasonal, unimodal discharge, but not necessarily as useful in bimodal-rainfall climates or in situations with relatively flat inflow profiles (e.g. in cases where a regulating dam is already present upstream).

Note that f_{reg} would normally be unity for the large (more-than-a-year storage) plants described in the previous sections; nevertheless, the user is free to specify a value smaller than unity for such plants when running REVUB. This could, for instance, reflect a requirement for a seasonal environmental flow, or a safeguard against extremely dry years to prevent regular operating rules from overdrawing the reservoir. In such a case, the operation of those plants, too, will follow what is described below, rather than what is described in the previous sections.

Once f_{reg} chosen, the fraction $Q_{in,frac}(t)$ is then assumed to be "storable" and, after accounting for the imposed environmental and irrigation outflow $Q_{stable}^{env/irr}(t)$, usable for balancing. REVUB assumes that the remainder of the flow, given by $Q_{in,nat}(t) - Q_{in,frac}(t) = (1 - f_{reg})Q_{in,nat}(t)$, passes through the reservoir without being stored, thus representing the seasonal "run-of-river" component $Q_{RoR}(t)$. Since f_{reg} can be calculated based on long-term average flow, such operation can be readily implemented as long as accurate measurements of inflow are available.

For $f_{reg} < 1$, d_{min} no longer represents the fraction of yearly average inflow required as minimum stable outflow, but the fraction of yearly average *storable* flow required as minimum stable outflow. The user must take this into account when specifying d_{min}. REVUB does this automatically when calculating a default d_{min} in case the user does not wish to prescribe a value[xiv].

The term Q_{RoR} is appended in equation (S3) (and would be an additional entry under the "water releases" arrows in Fig. S2):

$$\begin{aligned} V(t+\Delta t) &= V(t) + \left[Q_{in,nat}(t) - Q_{RoR}(t) - Q_{stable}(t) - Q_{flexible}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)]\,\frac{A(t)}{\rho}\right]\Delta t \\ &= V(t) + \left[Q_{in,frac}(t) - Q_{stable}(t) - Q_{flexible}(t) - Q_{spill}(t) + [\pi(t) - \lambda(t)]\,\frac{A(t)}{\rho}\right]\Delta t, \end{aligned} \tag{S31}$$

and the corresponding seasonal power generation is then equal to

$$P_{RoR}^{hydro}(t) = \min\left[Q_{RoR}(t); \max\left[0; Q_{turb}^{max} - Q_{stable}(t) - Q_{flexible}(t)\right]\right]\eta_{turb}\rho g h(t). \tag{S32}$$

Electricity generation from the hydropower component, denoted E_{RoR}^{hydro}, is then equal to

$$E_{RoR}^{hydro} = \int_{year} P_{RoR}^{hydro}(t)\ dt. \tag{S33}$$

Dispatching such as described for large hydropower plants in the previous sections is done similarly for small hydropower plants, with the following adaptations: (i) the component $Q_{in,frac}(t)$ is used as reference flow instead of $Q_{in,nat}(t)$ in equation (S4); (ii) the rule is added that, as soon as reservoir levels drop precariously low (in this case, below $f_{restart}V_{max}$), the RoR outflow (if nonzero) is temporarily stopped and the operation reverts to that of large hydropower plants, using the full inflow to stabilise water levels; and (iii) the seasonal power generation $P_{RoR}^{hydro}(t)$ is not included in equation (S9) as it is not expected to take part in the hour-to-hour load-following. Note, however, that if river discharge would be regulated by other reservoirs upstream in a cascade-like configuration, even $P_{RoR}^{hydro}(t)$ could become relatively stable throughout the year.

Note that, for extremely seasonal rivers, it is possible that the run-of-river component of discharge and power generation, after accounting for the stable and flexible components, fully "maxes out" the

[xiv] The calculation of this default, in a more generalised form for any f_{reg}, is as follows: $d_{min}^{default} = (f_{min}Q_{turb}^{max}/N_{turbines} - (1 - f_{reg})\min[Q_{in,nat}(t)])/(f_{reg}\overline{Q_{in,nat}(t)})$. The term $(1 - f_{reg})\min[Q_{in,nat}(t)]$ represents the fact that the run-of-river component can contribute to the minimum load of the first turbine provided that it has a nonzero minimum throughout the year. For $f_{reg} = 1$, which is the case for "large" hydropower plants, this equation reduces to $d_{min}^{default} = (f_{min}Q_{turb}^{max}/N_{turbines})/\overline{Q_{in,nat}(t)}$ as given earlier.

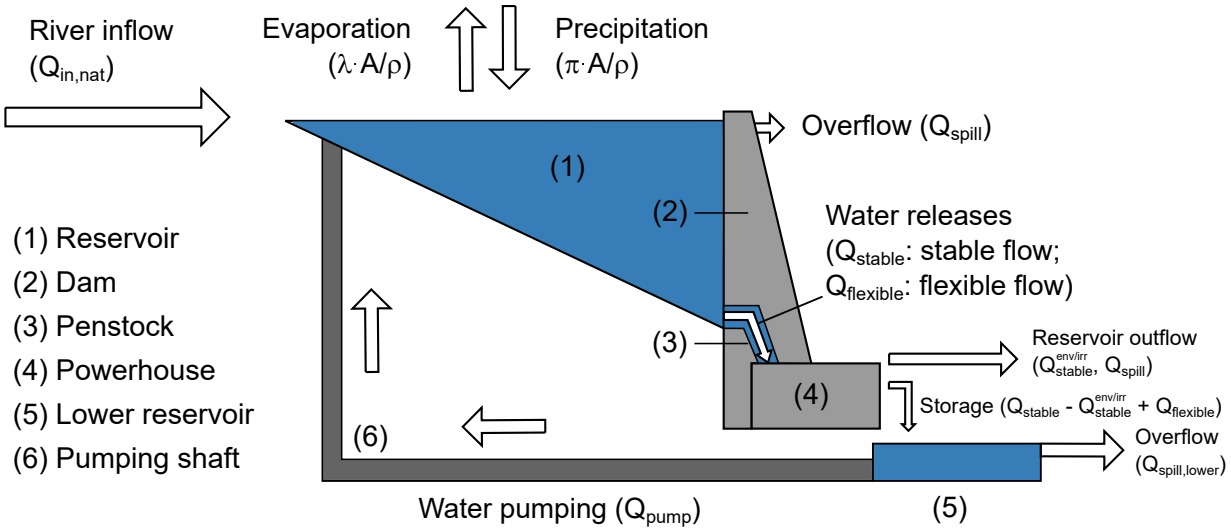


Figure S3: **Flowchart of the water gains and losses of a large hydropower plant, including pumped storage.** Water gains and losses of a hydropower plant modelled as pumped-storage plant, with an additional lower reservoir and a flow of water in the upstream direction for temporary energy storage.

turbine capacity. This would normally only happen during the periods in which the river discharge reaches its natural peak. In those periods, the plant automatically acts as baseload provider (using 100% of its discharge capacity and/or rated power capacity) and only reverts to the flexible operation once the inflow has somewhat reduced, freeing up operational reserves.

Note 5 Optional pumped-storage assessment

The REVUB code can also be used to assess the potential for increasing load-following potential by refurbishing large hydropower plants to pumped-storage schemes[19], such that excess production (typically of solar PV during mid-day) can be used to pump water back up into the reservoir, enabling the energy to be stored for use at a later stage. This would involve (i) building a downstream reservoir with volume $V_{lower,max} \ll V_{max}$ to store part of the released water, from which it can be pumped back up into the reservoir, and (ii) ensuring the turbines can be operated in reverse mode and/or pumps are installed.

The hydrological balance components then change as follows. First, since environmental and irrigation flow constraints still have to be met, the component $Q_{stable}^{env/irr}(t)$ has to be directly discharged downstream and cannot be used for filling the lower reservoir. Only the component $Q_{flexible}(t)$ and what remains of $Q_{stable}^{tech}(t)$, i.e. the part unaccounted for by $Q_{stable}^{env/irr}(t)$, can be used for filling the lower reservoir. Thus, the water balance of the lower reservoir is then given by

$$\begin{aligned} V_{lower}(t+\Delta t) &= V_{lower}(t) + \left[Q_{stable}(t) - Q_{stable}^{env/irr}(t) + Q_{flexible}(t) - Q_{pump}(t) - Q_{spill,lower}(t)\right]\Delta t \\ &= V_{lower}(t) + \left[\max\left[Q_{stable}^{tech}(t), Q_{stable}^{env/irr}(t)\right] - Q_{stable}^{env/irr}(t) + Q_{flexible}(t) - Q_{pump}(t) - Q_{spill,lower}(t)\right]\Delta t \\ &\equiv V_{lower}(t) + \left[Q_{in,lower}(t) - Q_{pump}(t) - Q_{spill,lower}(t)\right]\Delta t, \end{aligned} \tag{S34}$$

where $Q_{pump}(t)$ is the water pumped back up to the large reservoir for storage, $Q_{spill,lower}(t)$ is the overflow component of the lower reservoir, and $Q_{in,lower}(t)$ has been defined as the sum of the inflow into the lower reservoir from the upper reservoir for ease of notation. The precipitation and evaporation terms are neglected here as the lower reservoir is assumed to have a much smaller surface area than the upper reservoir. The water balance of the large reservoir is then given (cf. equation (S3)) by

$$V(t+\Delta t) = V(t) + \left[Q_{in,nat}(t) - Q_{stable}(t) - Q_{flexible}(t) + Q_{pump}(t) - Q_{spill}(t) + \left[\pi(t) - \lambda(t)\right]\frac{A(t)}{\rho}\right]\Delta t, \tag{S35}$$

and the net outflow of the pumped-storage plant is given (cf. equation (S2)) by

$$Q_{out}(t) = Q_{stable}^{env/irr}(t) + Q_{spill}(t) + Q_{spill,lower}(t). \tag{S36}$$

A schematic of this operation is shown in Fig. S3 (cf. Fig. S2). Note that REVUB only carries out pumped storage assessments for "large" hydropower plants; that is, those without run-of-river component. This is to prevent illogical solutions where pumping and release of RoR-flow could happen simultaneously. However, the user is free to set $f_{reg} = 1$ for any hydropower plant and override its "default" value, which suppresses any RoR component (and usually amplifies the drawdown-refill cycle). This allows to run a pumped-storage assessment also for plants that theoretically have filling times of less-than-a-year. As already touched upon in Note 4, this could be logical e.g. for plants with relatively unseasonal inflow.

The component $Q_{pump}(t)$ is calculated as follows. The power stored by pumping is given by

$$P_{pump}(t) = \left\{ \begin{array}{ll} 0, & \text{for } P_d(t) < 0 \\ \min\left[Q^{pot}_{pump}(t)\eta^{-1}_{pump}\rho g h(t); \min\left[P_d(t); P_{pump}(t-1) + \Delta P^{ramp}_{pump}\right]\right], & \text{for } P_d(t) \geq 0 \;\&\; \Delta P_d(t) \geq 0 \\ \min\left[Q^{pot}_{pump}(t)\eta^{-1}_{pump}\rho g h(t); \max\left[P_d(t); P_{pump}(t-1) - \Delta P^{ramp}_{pump}\right]\right], & \text{for } P_d(t) \geq 0 \;\&\; \Delta P_d(t) < 0 \end{array} \right\}, \quad \text{(S37)}$$

where ΔP^{ramp}_{pump} is the maximum ramp rate for pumping, η_{pump} is the pumping efficiency, and

$$Q^{pot}_{pump}(t) = \min\left[\frac{V_{lower}(t)}{\Delta t}; Q^{max}_{pump}\right] \quad \text{(S38)}$$

with Q^{max}_{pump} the maximum pump throughput. The flow pumped back into the reservoir then equals

$$Q_{pump}(t) = \frac{P_{pump}(t)}{\eta^{-1}_{pump}\rho g h(t)}. \quad \text{(S39)}$$

Finally, the overflow component of the lower reservoir is calculated as

$$Q_{spill,lower}(t) = \left\{ \begin{array}{ll} 0, & \text{for } \left(V_{lower,max} - V_{lower}(t)\right)/\Delta t \geq Q_{in,lower}(t) \\ Q_{in,lower}(t) - \left(V_{lower,max} - V_{lower}(t)\right)/\Delta t, & \text{for } \left(V_{lower,max} - V_{lower}(t)\right)/\Delta t < Q_{in,lower}(t) \end{array} \right\}. \quad \text{(S40)}$$

Since the pumping allows a certain fraction of the water to be recycled for power generation, the values c^{opt}_{solar} and c^{opt}_{wind} resulting from the optimisation described in Note 3.2 will end up higher as compared to a situation without pumping component. In this way, pumped storage can help increase the load that can be carried by the hydro-solar-wind combination.

The REVUB code includes an optional section to perform the above operations and the required optimisation (this scenario is named "STOR" in the code, as opposed to "CONV" and "BAL").

Note 6 Modelling cascade plants

The REVUB model includes provisions to allow modelling of cascaded plants, where two or more plants directly downstream of each other interact. In particular, REVUB allows for the following situations to be modelled:

- **A run-of-river-plant is located directly downstream of a reservoir plant.** In this case, the run-of-river plant needs only a very reduced amount of data input: installed capacity, hydraulic head, design discharge, number of turbines, and turbine efficiency. The upstream reservoir plant must be modelled with all the required parameters mentioned in the previous sections, and additionally it must be indicated that the upstream reservoir plant feeds the downstream one. In this way, the downstream plant will take the modelled outflow of the upstream one as inflow data, plus any unregulated lateral inflow. The CONV outflow from the upstream plant is used for this end[xv], multiplied with a boolean indicating whether the upstream reservoir plant was active in each hour (i.e. the assumption is that shutdowns upstream lead to shutdowns of the downstream plant too).

[xv] The logic for using CONV being that BAL outflow typically fluctuates strongly from hour to hour, but most run-of-river plants have some amount of "pondage" allowing to smoothen out such fluctuations.

- **A reservoir plant is located directly downstream of another reservoir plant; the upstream one is the main flexibility provider.** In this case, it is assumed that the operation of the upstream one is optimised based on its own reservoir capacity, and that the downstream one simply receives the resulting outflow of the upstream one and uses that for its own storage operation. The calculation is done as in the previous point, except that the downstream reservoir plant evidently needs the full set of parameters normally required for a REVUB simulation.

- **A reservoir plant is located directly downstream of another reservoir plant; the downstream one is the main flexibility provider.** In this case, it is assumed that the downstream plant uses the storage capacity of *both* reservoirs, denoted V_{down} (for the downstream plant) and V_{up} (for the upstream plant), respectively, to optimise its operation. Full sets of data for both plants must be entered, and the calculation is done as follows:

  1. The simplified assumption is made that any change ΔV in cumulatively stored volume $V_{cumul}(t) = V_{down}(t) + V_{up}(t)$ across the reservoirs is proportionally divided over both, according to shares f_{down} and f_{up} that the user can determine, with $f_{down} + f_{up} = 1$. In case of no specification by the user, this share automatically defaults to each reservoir's share in maximum total volume, i.e. $f_{down/up} = V^{max}_{down/up}/V^{max}_{cumul}$. Thus, if a total amount ΔV of previously stored water is turbined by the downstream plant, it is assumed that this results in a change in storage of $f_{down}\Delta V$ in the downstream reservoir, and $f_{up}\Delta V$ in the upstream reservoir.
  2. First, the simulation of the downstream reservoir plant is run. Any calculations involving overall storage volume, such as eq. (S3), are done using $V_{cumul}(t)$, and not $V_{down}(t)$, as basis. Based on the above logic, the amount of water stored in only the downstream reservoir is calculated as $V_{down}(t) = f_{down}V_{cumul}(t)$, and used to calculate the hydraulic head $h_{down}(t)$ at any point in time, needed to calculate power output of the downstream reservoir. For calculating evaporation and precipitation losses/gains, the surface area of both reservoirs is added to form $A_{cumul}(t) = A_{down}(t)+A_{up}(t)$, and any calculations involving overall area, such as again eq. (S3), are done with $A_{cumul}(t)$, and not $A_{down}(t)$, as basis. The regulation fraction $f_{reg,cumul}$, applied to operation of cumulative storage in the cascade, is interpreted to represent the fraction of total flow into the downstream reservoir ($Q_{in,down}(t)$) allocated for storage across the total available volume (V^{max}_{cumul})[xvi]. Lastly, the critical level of the "cumulative" storage is taken to be the maximum of $f_{stop,down}$ and $f_{stop,up}$ (and idem for $f_{restart}$).
  3. Second, the simulation of the upstream reservoir plant is run. Since its operation is dictated by the needs of the downstream plant (cf. the previous point), the calibration of its volume levels is slightly changed as compared to equation (S20): instead of using $V_{CONV}(t)$ as calibration series, we use the volume curve for the upper reservoir implied by the previous point, i.e. $V_{BAL,up}(t) = f_{up}V_{BAL,cumul}(t)$. The f_{reg} parameter is imposed on the upstream reservoir, based on the value of the same parameter for the cumulative storage (see above) and the ratio of respective average inflows: $f_{reg,up} = f_{reg,cumul} \cdot f_{up} \cdot \overline{Q_{in,down}}/\overline{Q_{in,up}}$.

Since the REVUB model does not currently have detailed routing procedures for cascades with reservoirs that lie relatively far apart, the application of the cascade module is currently limited to situations where downstream flow peak attenuation and flow delay between upstream and downstream reservoirs are small to negligible, i.e. where the reservoirs lie (more or less) directly downstream of one another.

Note 7 Note on spinning reserve requirements

In electricity networks, a certain reserve capacity, to be utilised in case of disruptions of supply, must always be available. The reserve capacity that is already online is denoted the spinning reserve,

[xvi] This assumes, implicitly, that the order of magnitude of any lateral flows feeding the downstream reservoir (i.e. flows which do not pass the upper reservoir) does not exceed the order of magnitude of the storage capacity of the downstream reservoir, which generally seems a reasonable assumption.

of which hydropower plants with reservoirs are typical providers. However, the flexible hydropower operation modeled by REVUB requires hydropower output to be increased during certain intervals to compensate for reductions in solar and/or wind power; thus, the remaining spinning reserves available to manage contingencies will consistently reach minimum values during those times.
The question is thus whether the hydropower plants could still respond to contingency events (e.g. failure of a transmission line leading to a solar power park temporarily being disconnected), despite being operated by flexible rule curves that lead to regular minima in the available spinning reserves. The spinning reserves $P_{spin}(t)$ from all hydropower plants within a certain geography can be calculated as

$$P_{spin}(t) = \sum_{\substack{\text{all hydro} \\ \text{plants}}} \left[1 - CF_{hydro}(t)\right] P^{r}_{turb} \Gamma(t), \tag{S41}$$

where $\Gamma(t)$ equals zero if the hydropower plant has temporarily undergone a drought-related shutdown, and unity otherwise; and $CF_{hydro}(t)$ is the capacity factor of each hydropower plant, calculated as

$$CF_{hydro}(t) = \frac{P^{hydro}_{stable}(t) + P^{hydro}_{flexible}(t) + P^{hydro}_{RoR}(t)}{P^{r}_{turb}}. \tag{S42}$$

To test the ability of hydropower to respond to supply disruptions of the added solar and wind power, the user could (ex-post) calculate the ratio (denoted ν) of $P_{spin}(t)$ to the total solar and wind power generation assumed to feed into the same grid. Typical benchmarks of spinning reserve needs for high-RE systems are in the order of $\nu \sim 10 - 20\%$ [22].

Note 8 Glossary

The table below provides an overview of all described parameters used in the REVUB model and its inputs and outputs. The symbol of each parameter is provided alongside its unit, a description of what it represents in the model, its data type (constant, variable, time series), its use in the model (as input, output, intermediate result, or used for preparing input), and its applicability across various components of the model. Here, "scenario" denotes a certain simulation setting: different scenarios may have different solar/wind mixes, different constraints on overproduction, different pools of hydropower plants contributing to flexibility, etc.

Symbol	Unit	Description	Type	Used as/for	Applicable to
$A(t)$	m^2	Reservoir lake surface area	Time series	Output	each hydropower plant & each scenario
A_{max}	m^2	Maximum reservoir lake surface area (reached at maximum filling level V_{max})	Constant	Input	each hydropower plant
$A_{down}(t)$	m^2	Estimated surface area of water of *downstream* reservoir in a cascade	Time series	Output	each cascade & each scenario
$A_{up}(t)$	m^2	Estimated surface area of water of *upstream* reservoir in a cascade	Time series	Output	each cascade & each scenario
$V_{cumul}(t)$	m^3	Cumulative surface area of water in reservoirs in a cascade. Equal to $A_{down}(t) + A_{up}(t)$	Time series	Output	each cascade & each scenario
C_{OR}	-	Operating reserve coefficient: Fraction of yearly reservoir inflow available for flexible use	Variable	-	each hydropower plant
C^{max}_{OR}	-	Maximum allowed operating reserve coefficient taking into account environmental flow needs	Constant	Input	each hydropower plant
C^{opt}_{OR}	-	Optimal operating reserve coefficient	Constant	Output	each hydropower plant & each scenario
c_{solar}	MW	Installed capacity of solar panels	Constant	Output	each hydropower plant & each scenario
c^{opt}_{solar}	MW	Optimal installed solar power capacity needed to follow optimal load $L_{opt}(t)$	Constant	Output	each hydropower plant & each scenario
c_{wind}	MW	Installed capacity of wind turbines	Constant	Output	each hydropower plant & each scenario
c^{opt}_{wind}	MW	Optimal installed wind power capacity needed to follow optimal load $L_{opt}(t)$	Constant	Output	each hydropower plant & each scenario
$CF_{hydro}(t)$	-	Capacity factor of hydropower plants	Time series	Output	each hydropower plant & each scenario
$CF_{solar}(t)$	-	Weighted average capacity factor of solar panels across locations	Time series	Input	each scenario
$CF^{n}_{solar}(t)$	-	Capacity factor of solar panels in each location with index n	Time series	Input	each solar power site

Symbol	Unit	Description	Type	Used as/for	Applicable to
$CF_{wind}(t)$	-	Weighted average capacity factor of wind turbines across locations	Time series	Input	each scenario
$CF^{n}_{wind}(t)$	-	Capacity factor of wind turbines in each location with index n	Time series	Input	each wind power site
d_{min}	-	Fraction of average storable reservoir inflow required as minimum outflow	Constant	Input	each hydropower plant
$d^{default}_{min}$	-	Default value of d_{min} used by REVUB in case of non-specification by user	Constant	Input	each hydropower plant
ΔP^{ramp}_{turb}	MW/min	Maximum ramp rate of hydropower plant	Constant	Input	each hydropower plant
ΔP^{ramp}_{pump}	MW/min	When simulating pumped-storage potential: Maximum ramp rate of pumps	Constant	Input	each large hydropower plant
$E^{hydro}_{reservoir}$	GWh/year	Total yearly hydropower generation	Yearly total	Output	each hydropower plant & each scenario
E^{hydro}_{stable}	GWh/year	Total yearly hydropower generation from stable reservoir outflow component	Yearly total	Output	each hydropower plant & each scenario
$E^{hydro}_{flexible}$	GWh/year	Total yearly hydropower generation from flexible reservoir outflow component	Yearly total	Output	each hydropower plant & each scenario
E^{hydro}_{RoR}	GWh/year	Total yearly hydropower generation from seasonal run-of-river outflow component	Yearly total	Output	each small hydropower plant & each scenario
E_{solar}	GWh/year	Total yearly solar power generation	Yearly total	Output	each hydropower plant & each scenario
E_{wind}	GWh/year	Total yearly wind power generation	Yearly total	Output	each hydropower plant & each scenario
ELCC	GWh/year	Effective Load Carrying Capability. Total yearly load followed by hydro-solar-wind without loss of load, ensuring long-term lake level stability, and meeting environmental flow requirements. Integral of $L_{followed}(t)$ over all time steps in a year.	Yearly total	Output	each hydropower plant & each scenario
ELCC_{tot}	GWh/year	Aggregate of ELCC across all power plants. Also called "total load-following potential" of hydro-solar-wind.	Yearly total	Output	each scenario
η_{turb}	-	Hydroturbine conversion efficiency (note: can be used to include representation of power factor)	Constant	Input	each hydropower plant
η_{pump}	-	When simulating pumped-storage potential: Pumping efficiency	Constant	Input	each hydropower plant
f_{down}	-	Fraction of downstream reservoir volume in cumulative storage volume in a cascade	Constant	Input	each cascade
f_{up}	-	Fraction of upstream reservoir volume in cumulative storage volume in a cascade	Constant	Input	each cascade
f_{opt}	-	Optimal filling fraction of hydropower reservoir	Constant	Input	each hydropower plant
f_{spill}	-	Reservoir filling fraction at which overflow prevention via spillways is initiated	Constant	Input	each hydropower plant
f_{stop}	-	Low reservoir filling fraction at which hydropower generation is curtailed	Constant	Input	each hydropower plant
$f_{restart}$	-	Reservoir filling fraction at which hydropower generation is restarted after curtailment	Constant	Input	each hydropower plant
f_{min}	-	Minimum stable turbine load to allow running in high-efficiency range	Constant	Input	each hydropower plant
f_{reg}	-	Fraction of yearly average natural inflow allocated to flexible use	Constant	Input	each hydropower plant
g	m/s^2	Gravitational acceleration	Constant	Input	general
γ	-	Conventional reservoir rule parameter	Constant	Input	each hydropower plant
$\Gamma(t)$	-	Binary function indicating whether hydropower plant is operating (1) or not (0)	Time series	Output	each hydropower plant & each scenario
$h(t)$	m	Hydraulic head: Difference in water level between headwater (behind the dam) and tailwater (at the turbines)	Time series	Output	each hydropower plant & each scenario
h_{max}	m	Maximum hydraulic head of hydropower scheme (reached at maximum filling level V_{max})	Constant	Input	each hydropower plant
$k_{turb}(t)$	-	Hydroturbine utilisation rate	Time series	Output	each hydropower plant & each scenario
κ	-	Conventional reservoir rule parameter	Constant	Input	each hydropower plant
$L(t)$	MW	Load to be followed	Time series	Intermediate	each scenario
$L_{opt}(t)$	MW	Optimal load to be followed by hydro-solar-wind mix, ensuring long-term reservoir lake level stability	Time series	Output	each hydropower plant & each scenario
$L_{followed}(t)$	MW	Actual followed load under optimal solution at zero loss of load (ideally equal to $L_{opt}(t)$)	Time series	Output	each hydropower plant & each scenario
$L_{res}(t)$	MW	Residual load: Difference between actual followed load and total hydro-solar-wind power generation	Time series	Intermediate	each hydropower plant & each scenario
$L_{norm}(t)$	-	Normalised load curve shape	Time series	Input	each scenario
$\lambda(t)$	$\text{kg/m}^2\text{/s}$	Evaporation flux on reservoir lake surface	Time series	Input	each hydropower plant

Symbol	Unit	Description	Type	Used as/for	Applicable to
μ	-	Constant used in modelling overflow prevention via spillways	Constant	Input	each hydropower plant
n	-	Index for different locations of solar and wind power generation	Index	-	solar power and wind power sites
N_{solar}	-	Total number of solar power generation sites connected to the same power grid	Constant	Pre-input	each scenario
N_{wind}	-	Total number of wind power generation sites connected to the same power grid	Constant	Pre-input	each scenario
$N_{turbines}$	-	Total number of turbines (units) of a single hydropower plant	Constant	Pre-input	each hydropower plant
$\nu(t)$	-	Ratio of spinning reserves to total solar and wind power generation	Time series	Output	each hydropower plant & each scenario
P^{r}_{turb}	MW	Rated hydroturbine capacity	Constant	Input	each hydropower plant
$P^{hydro}_{stable}(t)$	MW	Power generated by stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$P^{hydro}_{flexible}(t)$	MW	Power generated by flexible component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$P^{hydro}_{RoR}(t)$	MW	Power generated by seasonal run-of-river component of reservoir outflow	Time series	Output	each small hydropower plant & each scenario
$P_{inflexible}(t)$	MW	Inflexible power generation (stable hydro + solar + wind)	Time series	Intermediate	each hydropower plant & each scenario
$P_{solar}(t)$	MW	Solar power generation	Time series	Output	each hydropower plant & each scenario
$P_{wind}(t)$	MW	Wind power generation	Time series	Output	each hydropower plant & each scenario
$P_{followed}$	MW	Average actual followed load under optimal solution	Constant	Output	each hydropower plant & each scenario
$P^{xx}_{guaranteed}$	MW	Guaranteed power (based on user-defined exceedance percentile xx) across simulation horizon	Constant	Output	each hydropower plant & each scenario
P_{load}	MW	Average load to be followed	Constant	Intermediate	each hydropower plant & each scenario
$P_{spin}(t)$	MW	Spinning reserves available from all hydropower plants	Time series	Output	each hydropower plant & each scenario
$P_{pump}(t)$	MW	When simulating pumped-storage potential: Power used for pumping	Time series	Output	each large hydropower plant & each scenario
$P_{d}(t)$	MW	Difference between total inflexible power generation (stable hydro + solar + wind) and to-be-followed load	Time series	Output	each hydropower plant & each scenario
ϕ	-	Conventional reservoir rule parameter	Constant	Input	each hydropower plant
$\pi(t)$	kg/m^2/s	Precipitation flux on reservoir lake surface	Time series	Input	each hydropower plant
Ψ	-	Relative deviation between simulated lake levels under balancing-oriented and conventional operation, respectively	Function of c_{solar}, c_{wind}	Intermediate	each hydropower plant & each scenario
$Q_{in,frac}(t)$	m^3/s	Part of natural inflow assumed to be storable, equal to f_{reg} times the natural inflow	Time series	Intermediate	each small hydropower plant
$Q_{in,nat}(t)$	m^3/s	Natural river discharge into reservoir	Time series	Input	each hydropower plant
$Q_{in,lower}(t)$	m^3/s	Upper reservoir discharge into lower reservoir for pumped-storage plants, consisting of the non-environmental/irrigation component of stable outflow as well as flexible outflow	Time series	Input	each hydropower plant & each scenario
$Q_{out}(t)$	m^3/s	Total reservoir outflow, consisting of stable, flexible, run-of-river and spilling components	Time series	Output	each hydropower plant & each scenario
$Q_{stable}(t)$	m^3/s	Stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q^{tech}_{stable}(t)$	m^3/s	Technologically required part (related to minimum stable turbine load and need for active synchronous capacity) of stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q^{env/irr}_{stable}(t)$	m^3/s	Environmentally/irrigation-wise required part of stable component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q_{flexible}(t)$	m^3/s	Flexible component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
$Q_{RoR}(t)$	m^3/s	Seasonal run-of-river component of reservoir outflow (zero for large hydropower plants)	Time series	Output	each small hydropower plant
$Q_{spill}(t)$	m^3/s	Spilling component of reservoir outflow	Time series	Output	each hydropower plant & each scenario
Q^{max}_{turb}	m^3/s	Maximum hydroturbine throughput	Constant	Input	each hydropower plant
$Q^{pot}_{turb,flexible}(t)$	m^3/s	Maximum potential outflow available for flexible hydropower generation, before accounting for needs and ramping constraints	Time series	Intermediate	each hydropower plant & each scenario
$Q_{pump}(t)$	m^3/s	When simulating pumped-storage potential: Pumped flow	Time series	Output	each large hydropower plant & each scenario
Q^{max}_{pump}	m^3/s	When simulating pumped-storage potential: Maximum pumping throughput	Constant	Input	each large hydropower plant

Symbol	Unit	Description	Type	Used as/for	Applicable to
ρ	kg/m^3	Density of water	Constant	Input	general
t	hours	Time	Variable	-	general
T_{ref}	seconds	Number of seconds in a "reference period", e.g. a year	Constant	Input	general
τ_{fill}	-	Filling time: number of "reference periods" it takes (on average) to fill reservoir with natural inflow	Constant	Intermediate	each hydropower plant
$\tau_{fill,frac}$	-	Filling time: number of "reference periods" it takes (on average) to fill reservoir with fraction f_{reg} of natural inflow	Constant	Intermediate	each small hydropower plant
$V(t)$	m^3	Volume of water in reservoir	Time series	Output	each hydropower plant & each scenario
V_{max}	m^3	Maximum reservoir volume	Constant	Input	each hydropower plant
$V_{down}(t)$	m^3	Estimated volume of water in *downstream* reservoir in a cascade	Time series	Output	each cascade & each scenario
$V_{up}(t)$	m^3	Estimated volume of water in *upstream* reservoir in a cascade	Time series	Output	each cascade & each scenario
$V_{cumul}(t)$	m^3	Cumulative volume of water in reservoirs in a cascade. Equal to $V_{down}(t) + V_{up}(t)$	Time series	Output	each cascade & each scenario
$V_{lower}(t)$	m^3	When simulating pumped-storage potential: Volume of lower reservoir	Time series	Output	each large hydropower plant & each scenario
$V_{lower,max}$	m^3	When simulating pumped-storage potential: Maximum volume of lower reservoir	Constant	Input	each large hydropower plant
w^n_{solar}	-	Weight factor for capacity in each solar power site	Constant	Pre-input	each solar power site
w^n_{wind}	-	Weight factor for capacity in each wind power site	Constant	Pre-input	each wind power site
z_{max}	m	Maximal reservoir depth	Constant	Pre-input	each hydropower plant

Note 9 Running the model: Input (A)

The initialisation of REVUB simulations is done through the file **A_REVUB_initialise.py**, which reads data from the Excel file *control_file.xlsx* in which all parameters to be defined by the user can be controlled.

To learn how to practically use the control file to design REVUB simulations, the reader is referred to the Training dataset at `https://github.com/VUB-HYDR/REVUB/tree/master/4_Training`. Below, we provide a technical summary of the parameters that can be manipulated through the control file.

- Parameters that *differ for each hydropower plant* are set in the worksheet "`Hydropower plant parameters`" of the control file:
    - *Technical parameters*, which are constants specific to each hydropower plant. These include the installed capacity (P^r_{turb}), maximum turbine discharge (Q^{max}_{turb}), number of turbines ($N_{turbines}$), ramping rate (ΔP^{ramp}_{turb}), various volume fractions (f_{opt}, f_{spill}, f_{stop}, $f_{restart}$), turbine efficiency (η_{turb}), minimum stable turbine load (f_{min}), and multiple others. As it is possible to turn on/off individual hydropower plants for simulation inclusion (worksheet parameter `HPP_active`), the worksheet can simultaneously serve as overall database to collect hydropower plant data.
    - *Bathymetric relationships* for the modelled hydropower plants. These values should be read in by the code as an array with three columns and a user-determined number of rows; the first column should represent reservoir volume, and the second and third column the corresponding area and hydraulic head values. The relationships are set in the Excel file *data_bathymetry.xlsx* and linked to the control file through the parameter `HPP_name_data_bathymetry` in the worksheet "`Hydropower plant parameters`".
    - *Time series* of important parameters whose value needs to be known for each time step before a simulation can be performed, namely reservoir inflow, evaporation, precipitation, solar/wind power capacity factor, and the shape of the electricity demand profile. These values should be read in by the code as two-dimensional matrices for each hydropower

plant, with the following dimensions[xvii]: *[number of time slices (default: hours) per year; number of years covered by the simulation]*. These quantities are set in the other Excel files *data_xxx.xlsx* and linked to the control file through the parameters `HPP_name_data_xxx`.

- ◇ Note that hydrological time series (inflow, precipitation, evaporation) may often only be available at monthly timescale, not hourly. The repository contains a format converter code (**rearrange_data_to_hourly.py**) that parses user-specified monthly time series (to be entered in columns in the data collection file *rearrange_data_template.xlsx*, sheet `monthly_series`) into the matrix format required for REVUB. Running the converter code writes such a matrix at (pseudo-)hourly timescale back into *rearrange_data_template.xlsx*, from where it can then be copy-pasted into the relevant *data_xxx.xlsx* file.
- ◇ Similarly, time series for downstream irrigation needs may typically only be available at hourly (or different sub-daily) timescale for separate months of the year. The same may apply for e.g. simplified solar/wind CF input data in which only the basic sub-daily and seasonal dynamics are represented. The format converter **rearrange_data_to_hourly.py** can also handle this format. User-specified 24-h series for each month should be entered in columns in *rearrange_data_template.xlsx*, sheet `daily_bymonth_series`. Running the code then writes (pseudo)-hourly data back into *rearrange_data_template.xlsx*, to be copy-pasted into the relevant *data_xxx.xlsx* files.

- ○ *Indications* of which hydropower plant is in a cascade with which other one, either downstream or upstream. This is done through the parameters `HPP_cascade_upstream` (indicating the name of reservoir plants upstream) and `HPP_cascade_downstream` (indicating the name of reservoir or RoR plants downstream). The optional parameter `f_cascade_upstream` defines the share f_{up} explained in Note 6.
- ○ *Specification* of the solar/wind mix to be hybridised with each hydropower plant. The capacity ratio $c_{solar} : c_{wind}$ is quantified through the parameter `c_solar_relative`, which measures the relative fraction of solar in the solar-wind portfolio. (In mathematical terms, `c_solar_relative` $= c_{solar}/(c_{solar} + c_{wind})$.)
- ○ *Simulation accuracy parameters* specific to each hydropower plant. They are the following:
  - ◇ `f_init_BAL_start`, `f_init_BAL_step` and `f_init_BAL_end` determine the range (start, step size, and end) of $E_{solar} + E_{wind}$ (cf. equation (S25)), expressed as a fraction of $E^{hydro}_{reservoir,CONV}$ (cf. equation (S24)), i.e. the solution space in which the REVUB code starts searching for the solution to equation (S21). Note that the code gives a warning message to the user in case this range turns out to be inadequate (i.e. if no global minimum in Ψ is found). The same ranges should also be given for STOR scenarios (`f_init_STOR_start`, `f_init_STOR_step` and `f_init_STOR_end`) in case the user wishes to model these.
  - ◇ `N_refine_BAL` determines the accuracy with which the solution to equation (S21) is determined. After the initial search for this minimum in the range [`f_init_BAL_start` : `f_init_BAL_step` : `f_init_BAL_end`] (see previous point), the REVUB model can zoom in to the range around this minimum to identify its value with increased accuracy, using a step size reduced by a factor of ten. The number of times with which this is done is determined by `N_refine_BAL`. Thus, `N_refine_BAL` $= 1$ means that the initial search is deemed accurate enough, whereas each $+1$ in `N_refine_BAL` increases the accuracy of identification of the minimum in Ψ by one digit. Accordingly, each $+1$ also increases computation time proportionall `year_operation_start`

[xvii] Note that these technically refer to minimum dimensions. The user can, for instance, enter data for a higher amount of columns (years) than there are years in the simulation, and specify that REVUB should ignore those extra columns, using the worksheet "`General parameters`" of the control file (see below). This feature may be used to run initial tests using a limited number of years to speed up simulation time, before running with all years once satisfied with simulations settings.

and `year_operation_endy`. For users wishing to model STOR scenarios, the same principle applies to `N_refine_STOR`.

- ◇ In case the user already knows, from an earlier simulation, the optimal value in the range [`f_init_BAL_start` : `f_init_BAL_step` : `f_init_BAL_end`], they may enter it directly under the parameter `f_init_BAL_opt` to override the given range and speed up a simulation, which will run directly at that parameter instead of searching for the solution to equation (S21). The same can be done for STOR scenarios.
- ∗ The user can also, optionally, specify the calibration period (`year_calibration_start` and `year_calibration_end`) as well as the operation period (`year_operation_start` and `year_operation_end`) of each hydropower plant. More information on their use is given in Note 12.

- General modelling parameters *not specific to individual hydropower plants* are set on the worksheet "`General parameters`" of the control file. They allow to set the length of the time series to simulate; to specify which runs to include (CONV/BAL/STOR); to set physical, invariant quantities; and to set overall simulation accuracy parameters. Regarding the latter, the principal ones are given below:

  - `N_ELCC` determines the accuracy (the number of discrete steps between zero and maximum power generation) with which the zero-crossing of L_{res} in equation (S22) is determined.
  - `X_max` determines the amount of iterative loops employed by the model to get a converged estimate of $P^{hydro}_{stable,BAL/STOR}$ (cf. equation (S9) and the explanation below equation (S19)). Each +1 also increases computation time proportionally.
  - `psi_min_threshold` is zero by default, but can be set higher to speed up computation time in case the search for the minimum in Ψ takes too long. This parameter ensures that the zooming-in to higher accuracies (see previous point) is stopped as soon as $\min(\Psi)$ ends up below `psi_min_threshold`. It can thus be a compromise for users wishing to have high accuracy of determining $\min(\Psi)$ (i.e. setting `N_refine_BAL/STOR` > 1) but not at a high cost of computational speed.

Note 10 Running the model: Core code (B)

The actual core code of REVUB is contained in the file **B_REVUB_main_code.py**. Once the input data has been loaded using the file **A_REVUB_initialise.py**, the user can principally run this core code directly in the same namespace.

Note 11 Running the model: Output graphics and tables (C)

The results of the REVUB simulation are post-processed with two separate scripts, which generate several standardised figures and data tables based on the input and output of the model. The two scripts serve a somewhat different purpose:

- The script **C_REVUB_plotting_individual.py** is used to plot simulation results pertaining to a single (user-selected) hydropower plant, and produce data tables with hourly values of power generation, lake levels, and outflow for CONV/BAL/STOR runs of individual plants.
- The script **C_REVUB_plotting_multiple.py** is used to aggregate simulation results pertaining to (a user-selected set of) all hydropower plants. Based on a user-defined total electricity demand, it shows how much of this total demand can be met by hydro-solar-wind power on all involved timescales, and how much would remain to be met by other power sources. It plots the corresponding simulation results, as well as data tables with hourly values of power generation by source.

Note 11.1 Results for individual power plants

The figures generated by the file **C_REVUB_plotting_individual.py** are described below.

- Figure 1 shows (a) the (h, V) and (b) the (A, V) bathymetric calibration curves used as input to the model.
- Figure 2 shows (a) the full time series of hydraulic head $h(t)$ under CONV, BAL and (if modelled) STOR, (b) the frequency spectrum of these hydraulic head curves such that e.g. specific temporal signatures, such as diurnal dispatch in solar-heavy systems, can be discerned, and (c) the monthly median and interquartile ranges of natural inflow $Q_{in,nat}(t)$ and outflow $Q_{out}(t)$ (equation (S2)) under CONV, BAL and (if modelled) STOR.
- Figure 3 shows (a) the full time series of lake volume $V(t)$ under CONV, BAL and (if modelled) STOR, and (b) the monthly average time series of natural inflow $Q_{in,nat}(t)$ and outflow $Q_{out}(t)$ under CONV, BAL and (if modelled) STOR.
- Figure 4 shows the monthly average composition of the hydro-solar-wind mix supported by the selected hydropower plant, under the BAL scenario and for a user-defined year (see below). The figure indicates hydropower generation (by stable, flexible and RoR components) and the optimal solar and wind power contribution identified by REVUB. The achieved ELCC, whose profile reflects that of $L_{norm}(t)$, is also indicated. In case a STOR scenario was simulated, a corresponding extra figure will be produced for that scenario, additionally indicating the pump-stored part of solar/wind power generation.
- Figure 5 is analogous to Figure 4, but at yearly resolution and reflecting the entire simulated time series.
- Figure 6 is analogous to Figure 4, but at hourly (full) resolution for a user-defined time slice (see below).
- Figure 7 shows approximated reservoir release rules for a user-defined hour of the day during a user-defined month. The plot shows the median and interquartile range of needed reservoir release (minus the RoR and spill components, i.e. $Q_{stable}+Q_{flexible}$) versus the median hydraulic head, with each data point denoting results from one simulation year. A linear fit to these data points is also shown.
- Figure 8 shows statistics of turbine use over the simulated period. Based on the number of units installed, the amount of time in which a specific number of those is active is extracted from the hourly profiles of power generation and plotted in a bar chart.
- Figure 9 shows statistics of the operational regime, divided into six options defined at monthly scale: *flexibility* (the plant delivers flexibility throughout the month), *flexibility including pumping* (idem, but including recurrent pumping during that month [under STOR only]), *periodic saturation* (months in which the turbine capacity is maxed out during peak hours, and/or total outflow oscillates near the maximum discharge capacity of the turbines; see Note 4), *periodic saturation including pumping* (idem, but including recurrent pumping during that month [under STOR only]), *full saturation* (months in which the turbines are fully loaded, for instance during months with high RoR-components), and *shutdowns* (months in which droughts force hydropower curtailment).

To produce the figures, a few parameters need to be user-defined in the worksheets "`Plot power output (single HPP)`" of the file *control_file.xlsx*, where they can be changed to adapt the data to be plotted. These are as follows:

- `plot_HPP` is the index of the hydropower plant for which to plot results in all Figures.
- `plot_year` is the index of the simulation year for which to plot results in Figures 4 and 6.

- `plot_month` is the month in which to start the time slice in Figure 6.
- `plot_day_month` is the day of the month defined by `plot_month` on which to start the time slice in Figure 6.
- `plot_num_days` is the number of days for which to plot results in Figure 6.
- `plot_rules_month` is the month(s) of the year for which to plot results in Figure 7. This can be an array, e.g. [1, 4] would denote that the curves in Figure 7 are plotted both for January and for April.
- `plot_rules_hr` is the hour(s) of day during the month defined by `plot_rules_month` for which to plot results in Figure 7. This can be an array, e.g. [8, 20] would denote that the curves in Figure 7 are plotted both for 8 a.m. and 8 p.m.
- The user can further choose to include or exclude the RoR-component of hydropower in the graphs (`plot_RoR_part`), to include or exclude the plotting of the ELCC as a line (`plot_ELCC_line`), to turn the production of each individual figure on and off (`FigureX_on`), and to turn the production of data table output on and off (`tables_on`).

Note 11.2 Results for multiple power plants

The file **C_REVUB_plotting_multiple.py** serves to show to what extent the investigated hydropower, solar power and wind power plants could contribute to the overall power mix of a given territory. The principle of this file is as follows:

- To run the file, the user must first define several parameters to construct an overall electricity demand curve, denoted `P_total_hourly` (see below).
- The file then plots the aggregate power generation from all investigated hydro, solar and wind plants under BAL and STOR, and compares it to `P_total_hourly`.
- Any shortfalls in renewable power generation are then assumed to be filled up by thermal power plants, insofar as user-defined available thermal capacity permits. This reflects the assumption of a priority of dispatch for renewables. Any shortfalls unable to be met by this thermal capacity are assumed to be unsatisfied.
- When renewable power generation exceeds `P_total_hourly` by a certain amount, this amount is assumed to be curtailed.

In this way, the file allows to calculate the total power mix (hydro/solar/wind/thermal) suggested by the streamlined hydro-solar-wind operation as simulated by REVUB.
The figures produced by the file are described below.

- Figure 1 shows the monthly average composition of the hydro-solar-wind mix under the BAL scenario for a user-defined year (see below). The figure indicates the total power demand `P_total_hourly`, and how this power demand is met by the aggregate hydropower generation (by stable, flexible and RoR components) from all hydropower plants, the aggregate of all solar and wind power, and the necessary other (thermal) power generation. Any curtailed power (going beyond the user-defined overall demand) or unsatisfied power (unable to be met due to insufficient thermal capacity) is shown as such. In case a STOR scenario was simulated, a corresponding extra figure will be produced for that scenario, additionally indicating the pump-stored part of solar/wind power generation. (In this case, small hydropower plants for which no STOR scenario is available (cf. section Note 5) are included according to their BAL results.)
- Figure 2 is analogous to Figure 1, but at yearly resolution and reflecting the entire simulated time series.

- Figure 3 is analogous to Figure 1, but at hourly (full) resolution for a user-defined time slice (see below).
- Figure 4 shows average ramping envelopes of the hydro and thermal capacity across the same user-defined year as in Figure 1, for a user-defined range of hours. It contains two sub-figures, one of the ramping needs in absolute terms (left) and one of the ramping needs expressed as fraction of total capacity per hour (right).

To produce the figures, a few parameters need to be user-defined in the worksheet "`Plot power output (multi HPP)`" of the file *control_file.xlsx*, where they can be changed to adapt the data to be plotted. These are as follows:

- `plot_HPP_multiple` is an array containing the indices of the hydropower plants whose results are to be aggregated in the Figures. The user can thus select all, or a selection of, the simulated hydropower plants.
- `choose_demand_type` is a boolean operator, where `0` should be chosen to use the same (static) demand time series for every year, and `1` signifies the user wishes to show changes (dynamic) in annual demand levels.
- `P_total_av` and `P_r_total_thermal` represent the static values for average power demand and available thermal capacity, respectively, for the case `choose_demand_type = 0`.
- `P_total_av_series` and `P_r_total_thermal_series` represent the dynamic values (one for each year) for average power demand and available thermal capacity, respectively, for the case `choose_demand_type = 1`.
- `plot_year_multiple` is the index of the simulation year for which to plot results in Figures 1 and 3.
- `plot_month_multiple` is the month in which to start the time slice in Figure 3.
- `plot_day_month_multiple` is the day of the month defined by `plot_month` on which to start the time slice in Figure 3.
- `plot_num_days_multiple` is the number of days for which to plot results in Figure 3.
- `plot_ramping_range` is the number of hours across which ramping envelopes should be calculated.
- `chosen_load` is the name of the target load curve to be used in the plotting (found in the Excel file *data_load.xlsx*). This curve multiplied by `P_total_av_(series)` will constitute `P_total_hourly`.
- `plot_ELCC_line_multiple` allows the user to choose to include or exclude the plotting of the ELCC as a line.
- `tables_on_multiple` allows the user to choose to turn the production of output tables with hourly results on and off.

Note 12 Supplementary tips for model use

This section provides a few additional hints that may help to ease the use of the REVUB model in practice.

- As mentioned in sections Note 3.1 and Note 3.2, the default is to use the entire simulation period as reference for several processes in REVUB, such as the calculation and use of long-term average flow statistics (cf. eq. (S4), eq. (S14)) and the optimisation of lake level dynamics of BAL/STOR versus CONV (eq. (S20)). The user can change this by manually setting the

parameters `year_calibration_start` and `year_calibration_end` in the sheet "`Hydropower plant parameters`" of the control file *control_file.xlsx*. This can be useful, for instance, when simulating extreme events such as unforeseen droughts hitting a hydropower plant: in that case, one would select the calibration period to not include that extreme year, forcing the model to be "myopic" in relation to the extreme dry period[4]. Leaving the parameter fields empty will revert the model to the default of using the entire simulation period.

- Similarly, the user can specify that a hydropower plant enters into operation (and/or is taken out of operation) in a specific year of the modelled period, by manually setting the parameters `year_operation_start` and `year_operation_end` in the sheet "`Hydropower plant parameters`" of the control file *control_file.xlsx*. These years do not need to coincide, or overlap at all, with the calibration period mentioned in the above point. In case the calibration period includes years that are not part of the operation period, the code output will provide water-related parameters (reservoir filling levels, flows, ...) for all years spanned by calibration and operation period (allowing to see the calibration results), but energy-related parameters only for the operation period (as the plant is assumed unoperational for other years). Leaving the parameter fields empty will revert the model to the default of using the entire simulation period.

- In order to check several basic outcomes of the model (such as lake levels, outflow statistics, etc.), it is instructive to first run a CONV scenario without bothering with the BAL/STOR optimisations, allowing to easily identify obvious errors in the input data. For this reason, it is recommended to first run a "calibration" scenario with only CONV by setting `calibration_only` to zero in the worksheet "`General parameters`" in the control file. This will skip the BAL/STOR optimisation entirely. In that case, it is not necessary to specify `HPP_name_data_load`.

- One could conceivably wish to investigate how a hydropower plant performs flexibly (BAL) compared to baseload-like (CONV), but *without* any contribution from solar and/or wind power, i.e. solely to meet a certain target load without hybridisation with other power sources. This can be simulated by simply not linking any solar and wind timeseries to the simulation (leaving `HPP_name_data_CF_solar` and `HPP_name_data_CF_wind` empty in the sheet "`Hydropower plant parameters`" of the control file), in which case REVUB will run the simulation without solar and wind. In that case, the code uses an alternative for eq. (S11), namely

$$P_{load} = P^r_{turb} \cdot f_{reg} \cdot c_{dummy}, \tag{S43}$$

  with c_{dummy} a dummy variable analogous to c_{solar} and c_{wind} from eq. (S9), so that eq. (S21) becomes

$$\min(\Psi) = \Psi(c^{opt}_{dummy}). \tag{S44}$$

- For simplicity, if the user wishes to ignore precipitation and evaporation effects in the simulation (usually second-order effects), they can simply choose to not link any precipitation or evaporation time series to the hydropower plants in question. This is done in the sheet "`Hydropower plant parameters`" of the control file by leaving the parameters `HPP_name_data_precipitation` and `HPP_name_data_evaporation` empty. The same goes for the prescribed outflow, through the parameter `HPP_name_data_outflow_prescribed`.

- One could simulate a "prudent management scheme" (i.e. deliberately keeping more water stored in the reservoir than the REVUB optimisation would suggest, for instance to increase resilience to dry years) by setting the parameter `f_init_BAL_opt` deliberately somewhat below its actual optimal value.

- Although REVUB is primarily designed to simulate reservoir hydropower plants, it can be used in "simplified" mode to simulate run-of-river plants. In this case, most of the input parameters can be ignored as they are not relevant to plants without reservoirs. The only remaining parameters of importance are the inflow (`HPP_name_data_inflow` in the sheet "`Hydropower`

748 `plant parameters`" of the control file, linking to the correct inflow time series), rated ca-
749 pacity (`P_r_turb`), hydraulic head (`h_max`), design discharge (`Q_max_turb`), number of turbines
750 (`no_turbines`), and turbine efficiency (`eta_turb`). All other fields in the sheet "`Hydropower`
751 `plant parameters`" can be left empty. In case the run-of-river plant is part of a cascade with
752 reservoir plants upstream, even the inflow parameter can be left empty as the plant will receive
753 the outflow of the upstream plant as inflow (cf. Note 6).

References

[1] Sterl, S. *et al.* Smart renewable electricity portfolios in West Africa. *Nature Sustainability* **3**, 710–719 (2020).

[2] Sterl, S., Donk, P., Willems, P. & Thiery, W. Turbines of the Caribbean: Decarbonising Suriname's electricity mix through hydro-supported integration of wind power. *Renewable and Sustainable Energy Reviews* **134**, 110352 (2020).

[3] Sterl, S., Fadly, D., Liersch, S., Koch, H. & Thiery, W. Linking solar and wind power in eastern Africa with operation of the Grand Ethiopian Renaissance Dam. *Nature Energy* **6**, 407–418 (2021).

[4] Sterl, S. *et al.* Variable renewables fortify Ecuador's power system against recurrences of drought-driven energy crises. *Nature Water* **4**, 571–585 (2026).

[5] Liersch, S. *et al.* Vulnerability of rice production in the Inner Niger Delta to water resources management under climate variability and change. *Environmental Science and Policy* **34**, 18 – 33 (2013).

[6] Oyerinde, G. T. *et al.* Quantifying uncertainties in modeling climate change impacts on hydropower production. *Climate* **4** (2016).

[7] Vanderkelen, I., van Lipzig, N. P. M. & Thiery, W. Modelling the water balance of Lake Victoria (East Africa) - Part 1: Observational analysis. *Hydrology and Earth System Sciences* **22**, 5509–5525 (2018).

[8] Liersch, S. *et al.* Water resources planning in the Upper Niger River basin: Are there gaps between water demand and supply? *Journal of Hydrology: Regional Studies* **21**, 176 – 194 (2019).

[9] IRENA. Power System Flexibility for the Energy Transition, Part 1: Overview for Policy Makers (2018). URL `https://www.irena.org/publications/2018/Nov/Power-system-flexibility-for-the-energy-transition`. International Renewable Energy Agency, Abu Dhabi.

[10] Jägermeyr, J., Pastor, A., Biemans, H. & Gerten, D. Reconciling irrigated food production with environmental flows for Sustainable Development Goals implementation. *Nature Communications* **8**, 15900 (2017).

[11] Bakken, T. H., Killingtveit, Å., Engeland, K., Alfredsen, K. & Harby, A. Water consumption from hydropower plants - review of published estimates and an assessment of the concept. *Hydrology and Earth System Sciences* **17**, 3983–4000 (2013).

[12] Proussevitch, A. *et al.* Log-exponential reservoir operating rules for global and regional hydrological modeling (2013). URL `https://scholars.unh.edu/earthsci_facpub/383/`. Abstract GC21B-0827 presented at 2013 Fall Meeting, AGU, San Francisco, 9-13 December 2013.

[13] Yassin, F. *et al.* Representation and improved parameterization of reservoir operation in hydrological and land-surface models. *Hydrology and Earth System Sciences* **23**, 3735–3764 (2019).

[14] Engeland, K. *et al.* Space-time variability of climate variables and intermittent renewable electricity production - A review. *Renewable and Sustainable Energy Reviews* **79**, 600 – 617 (2017).

[15] Yang, W. *et al.* Burden on hydropower units for short-term balancing of renewable power systems. *Nature Communications* **9**, 2633 (2018).

[16] Sterl, S. *et al.* An all-Africa dataset of energy model "supply regions" for solar photovoltaic and wind power. *Scientific Data* **9**, 664 (2022).

[17] Sterl, S. *et al.* Smart renewable electricity portfolios in West Africa. *Nature Sustainability* **3**, 710–719 (2020).

[18] Homer Energy. HOMER Pro 3.11 - operating reserve (n.d.). URL `https://www.homerenergy.com/products/pro/docs/3.11/operating_reserve.html`.

[19] Jurasz, J., Mikulik, J., Krzywda, M., Ciapała, B. & Janowski, M. Integrating a wind- and solar-powered hybrid to the power system by coupling it with a hydroelectric power station with pumping installation. *Energy* **144**, 549 – 563 (2018).

[20] Kaveh, K., Hosseinjanzadeh, H. & Hosseini, K. A new equation for calculation of reservoir's area-capacity curves. *KSCE Journal of Civil Engineering* **17**, 1149–1156 (2013).

[21] Turner, S. W., Ng, J. Y. & Galelli, S. Examining global electricity supply vulnerability to climate change using a high-fidelity hydropower dam model. *Science of The Total Environment* **590-591**, 663 – 675 (2017).

[22] Ortega-Vazquez, M. A. & Kirschen, D. M. Estimating the Spinning Reserve Requirements in Systems With Significant Wind Power Generation Penetration. *IEEE Transactions on Power Systems* **24**, 114–124 (2009).